Parallel Seam Carving using OpenMP

Assignment 1 — High Performance Computing

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Exercise 1: We parallelize all algorithm steps with OpenMP, specifically using the basic parallel approach for cumulative energy and parallelizing only the copying phase of seam removal. All code is compiled with `-O3` optimizations.

Execution times are measured across five image resolutions (720×480 to 7680×4320) using 1, 2, 4, 8, 16, and 32 threads. Each test case requires at least 5 runs removing 128 seams to calculate average timings. Performance is evaluated via speed-up, defined as $S = \frac{t_s}{t_p}$, where t_s is the sequential runtime and t_p is the parallel runtime.

Exercise 2: We implement greedy seam removal to enable the parallel processing of multiple seams and we analyze the trade-offs between execution speed and final image quality. For each algorithm step and image resolution, we perform a scaling study to determine the optimal thread and core counts.

We enhance the energy calculation phase by implementing the improved triangle-based cumulative energy approach. Furthermore, we further improve the algorithm by running the algorithm from top and bottom and the same time and joining the values in the middle.

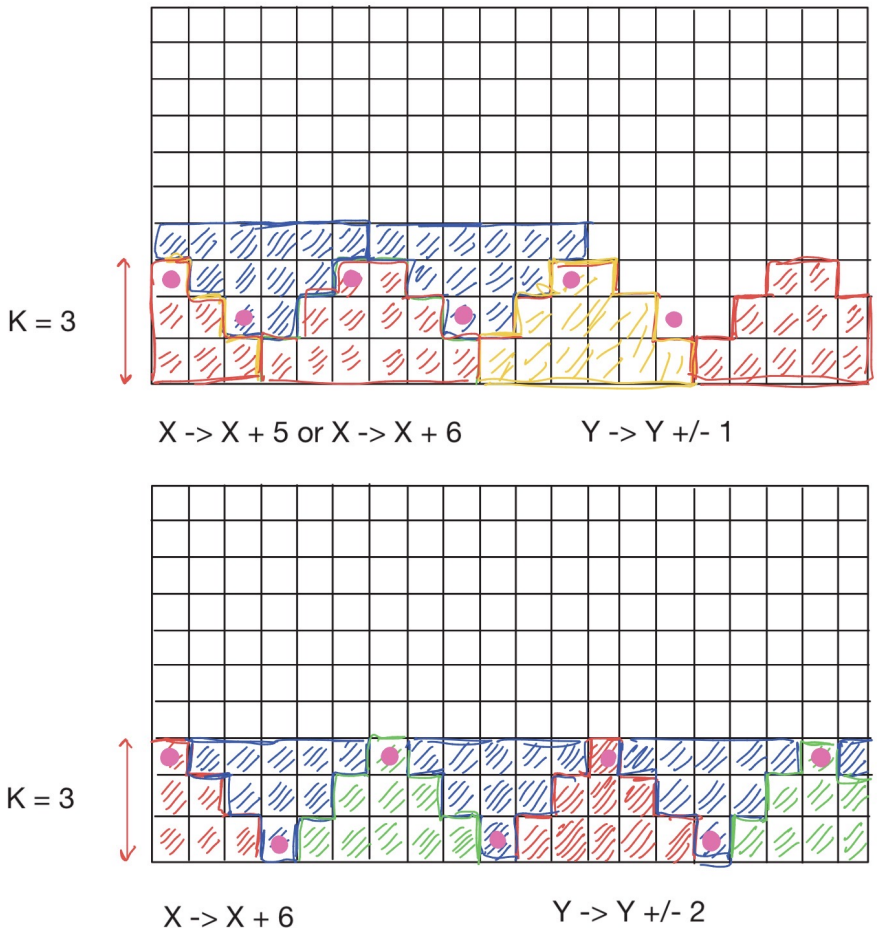

Figure 1: This is the description that appears under your image.